

38—(MathAA/10_HL_Winter_2022_Q6)—Probability

[Maximum mark: 6]

Events A and B are such that $P(A) = 0.3$ and $P(B) = 0.8$.

- (a) Determine the value of $P(A \cap B)$ in the case where the events A and B are independent. [1]
- (b) Determine the minimum possible value of $P(A \cap B)$. [3]
- (c) Determine the maximum possible value of $P(A \cap B)$, justifying your answer. [2]



1-(MAT/21_HL_Summer_2012_Q1)-Differentiation

[Maximum mark: 5]

Given that the graph of $y = x^3 - 6x^2 + kx - 4$ has exactly one point at which the gradient is zero, find the value of k .

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



2-(MAT/22_HL_Summer_2012_Q6)-Differentiation

- (a) Sketch the curve $y = \frac{\cos x}{\sqrt{x^2 + 1}}$, $-4 \leq x \leq 4$ showing clearly the coordinates of the x -intercepts, any maximum points and any minimum points. [4 marks]

- (b) Write down the gradient of the curve at $x = 1$. [1 mark]
- (c) Find the equation of the normal to the curve at $x = 1$. [3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....



3-(MAT/21_HL_Summer_2012_Q10)-Differentiation

[Maximum mark: 8]

A triangle is formed by the three lines $y = 10 - 2x$, $y = mx$ and $y = -\frac{1}{m}x$, where $m > \frac{1}{2}$. Find the value of m for which the area of the triangle is a minimum.



4-(MAT/21_HL_Summer_2012_Q11)-Differentiation

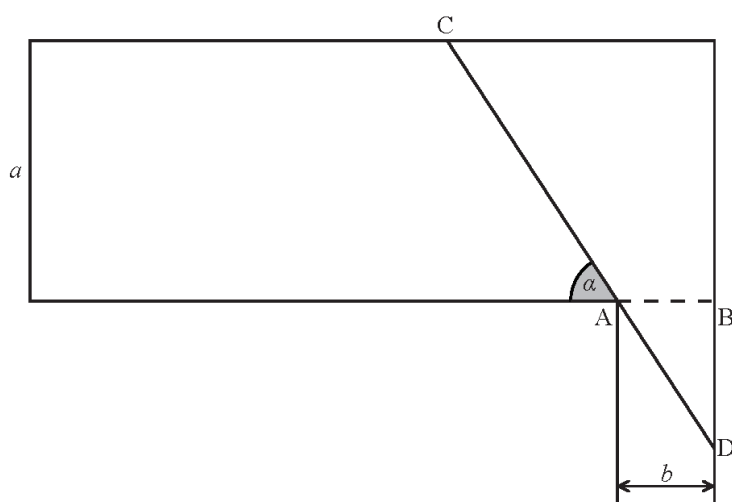
The function $f(x) = 3 \sin x + 4 \cos x$ is defined for $0 < x < 2\pi$.

- (a) Write down the coordinates of the minimum point on the graph of f . [1 mark]
- (b) The points $P(p, 3)$ and $Q(q, 3)$, $q > p$, lie on the graph of $y = f(x)$. Find p and q . [2 marks]
- (c) Find the coordinates of the point, on $y = f(x)$, where the gradient of the graph is 3. [4 marks]
- (d) Find the coordinates of the point of intersection of the normals to the graph at the points P and Q. [7 marks]



5-(MAT/20_HL_Winter_2012_Q12)-Differentiation

The diagram shows the plan of an art gallery a metres wide. $[AB]$ represents a doorway, leading to an exit corridor b metres wide. In order to remove a painting from the art gallery, CD (denoted by L) is measured for various values of α , as represented in the diagram.



- (a) If α is the angle between $[CD]$ and the wall, show that $L = \frac{a}{\sin \alpha} + \frac{b}{\cos \alpha}$,
 $0 < \alpha < \frac{\pi}{2}$. [3 marks]

- (b) If $a = 5$ and $b = 1$, find the maximum length of a painting that can be removed through this doorway. [4 marks]

Let $a = 3k$ and $b = k$.

- (c) Find $\frac{dL}{d\alpha}$. [3 marks]

- (d) Find, in terms of k , the maximum length of a painting that can be removed from the gallery through this doorway. [6 marks]

- (e) Find the minimum value of k if a painting 8 metres long is to be removed through this doorway. [2 marks]



6-(MAT/22_HL_Summer_2013_Q12)-Differentiation,Integration

Consider the differential equation $y \frac{dy}{dx} = \cos 2x$.

(a) (i) Show that the function $y = \cos x + \sin x$ satisfies the differential equation.

(ii) Find the general solution of the differential equation. Express your solution in the form $y = f(x)$, involving a constant of integration.

(iii) For which value of the constant of integration does your solution coincide with the function given in part (i)?

[10 marks]

(b) A different solution of the differential equation, satisfying $y = 2$ when $x = \frac{\pi}{4}$, defines a curve C .

(i) Determine the equation of C in the form $y = g(x)$, and state the range of the function g .

A region R in the xy plane is bounded by C , the x -axis and the vertical lines $x = 0$ and $x = \frac{\pi}{2}$.

(ii) Find the area of R .

(iii) Find the volume generated when that part of R above the line $y = 1$ is rotated about the x -axis through 2π radians.

[12 marks]



7-(MAT/21_HL_Summer_2013_Q13)-Differentiation

The function f has inverse f^{-1} and derivative $f'(x)$ for all $x \in \mathbb{R}$. For all functions with these properties you are given the result that for $a \in \mathbb{R}$ with $b = f(a)$ and $f'(a) \neq 0$

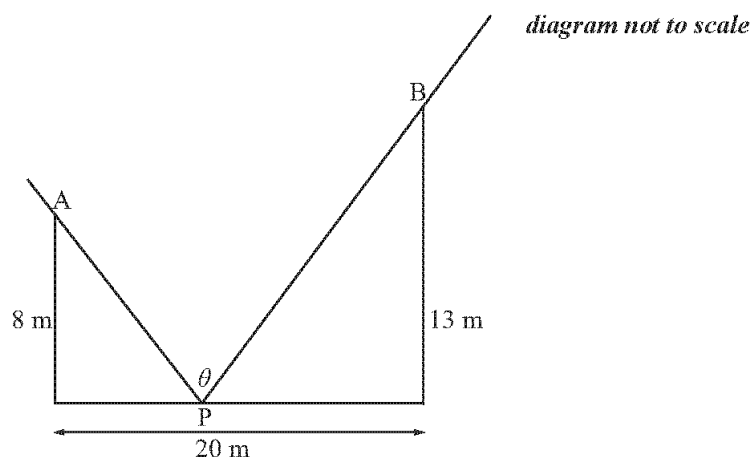
$$(f^{-1})'(b) = \frac{1}{f'(a)}.$$

- (a) Verify that this is true for $f(x) = x^3 + 1$ at $x = 2$. [6 marks]
- (b) Given that $g(x) = xe^{x^2}$, show that $g'(x) > 0$ for all values of x . [3 marks]
- (c) Using the result given at the start of the question, find the value of the gradient function of $y = g^{-1}(x)$ at $x = 2$. [4 marks]
- (d) (i) With f and g as defined in parts (a) and (b), solve $g \circ f(x) = 2$.
(ii) Let $h(x) = (g \circ f)^{-1}(x)$. Find $h'(2)$. [6 marks]



8-(MAT/22_HL_Summer_2013_Q13)-Differentiation

A straight street of width 20 metres is bounded on its parallel sides by two vertical walls, one of height 13 metres, the other of height 8 metres. The intensity of light at point P at ground level on the street is proportional to the angle θ where $\theta = \hat{APB}$, as shown in the diagram.



- (a) Find an expression for θ in terms of x , where x is the distance of P from the base of the wall of height 8 m. [2 marks]
- (b) (i) Calculate the value of θ when $x = 0$.
(ii) Calculate the value of θ when $x = 20$. [2 marks]
- (c) Sketch the graph of θ , for $0 \leq x \leq 20$. [2 marks]
- (d) Show that $\frac{d\theta}{dx} = \frac{5(744 - 64x - x^2)}{(x^2 + 64)(x^2 - 40x + 569)}$. [6 marks]
- (e) Using the result in part (d), or otherwise, determine the value of x corresponding to the maximum light intensity at P. Give your answer to four significant figures. [3 marks]
- (f) The point P moves across the street with speed 0.5 m s^{-1} . Determine the rate of change of θ with respect to time when P is at the midpoint of the street. [4 marks]



9-(MAT/22_HL_Summer_2014_Q9)-Differentiation

[Maximum mark: 5]

Sand is being poured to form a cone of height h cm and base radius r cm. The height remains equal to the base radius at all times. The height of the cone is increasing at a rate of 0.5 cm min^{-1} .

Find the rate at which sand is being poured, in $\text{cm}^3 \text{ min}^{-1}$, when the height is 4 cm.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



10–(MAT/21_HL_Summer_2014_Q10)–Functions-Roots,Differentiation

Let $f(x) = \frac{e^{2x} + 1}{e^x - 2}$.

(a) Find the equations of the horizontal and vertical asymptotes of the curve $y = f(x)$. [4]

(b) (i) Find $f'(x)$.

(ii) Show that the curve has exactly one point where its tangent is horizontal.

(iii) Find the coordinates of this point. [8]

(c) Find the equation of L_1 , the normal to the curve at the point where it crosses the y -axis. [4]

The line L_2 is parallel to L_1 and tangent to the curve $y = f(x)$.

(d) Find the equation of the line L_2 . [5]



11-(MAT/22_HL_Summer_2014_Q10)-Differentiation

Consider the curve with equation $(x^2 + y^2)^2 = 4xy^2$.

- (a) Use implicit differentiation to find an expression for $\frac{dy}{dx}$. [5]
- (b) Find the equation of the normal to the curve at the point (1, 1). [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

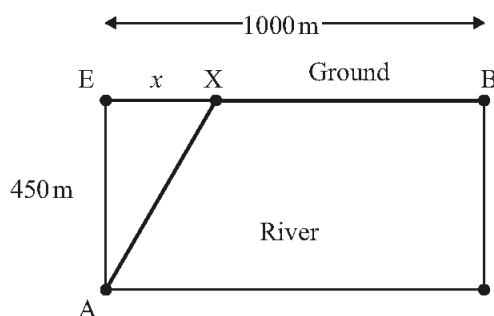
exam-mate.co



12-(MAT/22_HL_Summer_2014_Q12)-Differentiation

Engineers need to lay pipes to connect two cities A and B that are separated by a river of width 450 metres as shown in the following diagram. They plan to lay the pipes under the river from A to X and then under the ground from X to B. The cost of laying the pipes under the river is five times the cost of laying the pipes under the ground.

Let $EX = x$.



Let k be the cost, in dollars per metre, of laying the pipes under the ground.

- (a) Show that the total cost C , in dollars, of laying the pipes from A to B is given by $C = 5k\sqrt{202500 + x^2} + (1000 - x)k$. [2]

- (b) (i) Find $\frac{dC}{dx}$.

- (ii) Hence find the value of x for which the total cost is a minimum, justifying that this value is a minimum. [7]

- (c) Find the minimum total cost in terms of k . [1]

The angle at which the pipes are joined is $\hat{AXB} = \theta$.

- (d) Find θ for the value of x calculated in (b). [2]

For safety reasons θ must be at least 120° .

Given this new requirement,

- (e) (i) find the new value of x which minimises the total cost;
(ii) find the percentage increase in the minimum total cost. [3]



13-(MAT/20_HL_Winter_2014_Q10)-Differentiation

Consider the triangle PQR where $\hat{QPR} = 30^\circ$, $PQ = (x+2)\text{ cm}$ and $PR = (5-x)^2\text{ cm}$, where $-2 < x < 5$.

(a) Show that the area, $A\text{ cm}^2$, of the triangle is given by $A = \frac{1}{4}(x^3 - 8x^2 + 5x + 50)$. [2]

(b) (i) State $\frac{dA}{dx}$.

(ii) Verify that $\frac{dA}{dx} = 0$ when $x = \frac{1}{3}$. [3]

(c) (i) Find $\frac{d^2A}{dx^2}$ and hence justify that $x = \frac{1}{3}$ gives the maximum area of triangle PQR.

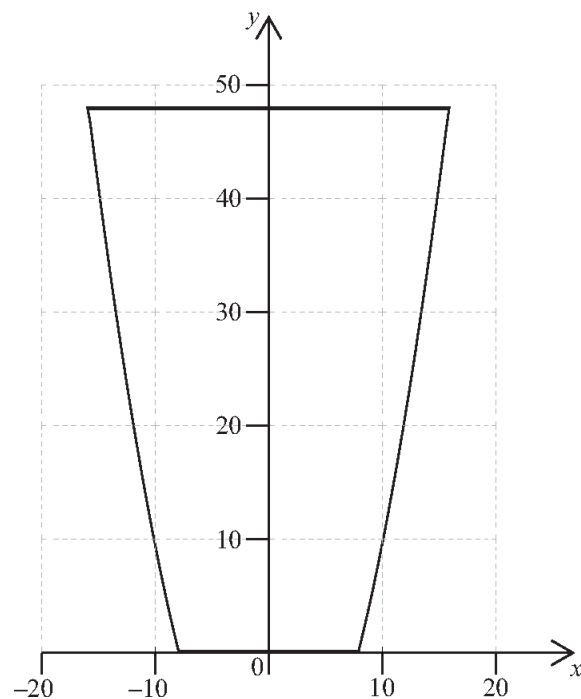
(ii) State the maximum area of triangle PQR.

(iii) Find QR when the area of triangle PQR is a maximum. [7]



14-(MAT/20_HL_Winter_2014_Q13)-Differentiation, Integration

The vertical cross-section of a container is shown in the following diagram.



The curved sides of the cross-section are given by the equation $y = 0.25x^2 - 16$. The horizontal cross-sections are circular. The depth of the container is 48 cm.

- (a) If the container is filled with water to a depth of h cm, show that the volume, V cm³, of the water is given by $V = 4\pi \left(\frac{h^2}{2} + 16h \right)$. [3]



The container, initially full of water, begins leaking from a small hole at a rate given by $\frac{dV}{dt} = -\frac{250\sqrt{h}}{\pi(h+16)}$ where t is measured in seconds.

(b) (i) Show that $\frac{dh}{dt} = -\frac{250\sqrt{h}}{4\pi^2(h+16)^2}$.

(ii) State $\frac{dt}{dh}$ and hence show that $t = \frac{-4\pi^2}{250} \int \left(h^{\frac{3}{2}} + 32h^{\frac{1}{2}} + 256h^{-\frac{1}{2}} \right) dh$.

(iii) Find, correct to the nearest minute, the time taken for the container to become empty. (60 seconds = 1 minute) [10]

Once empty, water is pumped back into the container at a rate of $8.5 \text{ cm}^3 \text{ s}^{-1}$. At the same time, water continues leaking from the container at a rate of $\frac{250\sqrt{h}}{\pi(h+16)} \text{ cm}^3 \text{ s}^{-1}$.

(c) Using an appropriate sketch graph, determine the depth at which the water ultimately stabilizes in the container. [3]



15-(MAT/21_HL_Summer_2015_Q5)-Differentiation

[Maximum mark: 5]

A bicycle inner tube can be considered as a joined up cylinder of fixed length 200 cm and radius r cm. The radius r increases as the inner tube is pumped up. Air is being pumped into the inner tube so that the volume of air in the tube increases at a constant rate of $30 \text{ cm}^3 \text{ s}^{-1}$. Find the rate at which the radius of the inner tube is increasing when $r = 2 \text{ cm}$.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



16-(MAT/21_HL_Summer_2015_Q6)-Differentiation

[Maximum mark: 4]

A function f is defined by $f(x) = x^3 + e^x + 1$, $x \in \mathbb{R}$. By considering $f'(x)$ determine whether f is a one-to-one or a many-to-one function.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



17-(MAT/21_HL_Summer_2015_Q9)-Differentiation

[Maximum mark: 7]

Find the equation of the normal to the curve $y = \frac{e^x \cos x \ln(x+e)}{(x^{17}+1)^5}$ at the point where $x = 0$.

In your answer give the value of the gradient, of the normal, to three decimal places.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



18-(MAT/22_HL_Summer_2015_Q11)-Differentiation

A curve is defined by $x^2 - 5xy + y^2 = 7$.

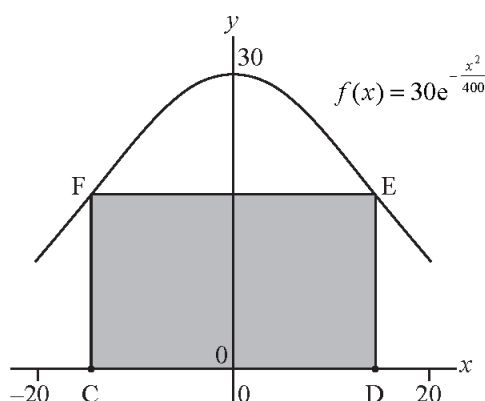
- (a) Show that $\frac{dy}{dx} = \frac{5y - 2x}{2y - 5x}$. [3]
- (b) Find the equation of the normal to the curve at the point $(6, 1)$. [4]
- (c) Find the distance between the two points on the curve where each tangent is parallel to the line $y = x$. [8]



19-(MAT/20_HL_Winter_2015_Q13)-Differentiation

The following diagram shows a vertical cross section of a building. The cross section of the roof of the building can be modelled by the curve $f(x) = 30e^{-\frac{x^2}{400}}$, where $-20 \leq x \leq 20$.

Ground level is represented by the x -axis.



(a) Find $f''(x)$. [4]

(b) Show that the gradient of the roof function is greatest when $x = -\sqrt{200}$. [3]

The cross section of the living space under the roof can be modelled by a rectangle CDEF with points $C(-a, 0)$ and $D(a, 0)$, where $0 < a \leq 20$.

(c) Show that the maximum area A of the rectangle CDEF is $600\sqrt{2}e^{-\frac{1}{2}}$. [5]

(d) A function I is known as the Insulation Factor of CDEF. The function is defined as $I(a) = \frac{P(a)}{A(a)}$ where P = Perimeter and A = Area of the rectangle.

(i) Find an expression for P in terms of a .

(ii) Find the value of a which minimizes I .

(iii) Using the value of a found in part (ii) calculate the percentage of the cross sectional area under the whole roof that is not included in the cross section of the living space. [9]



20–(MAT/22_HL_Summer_2016_Q7)–Differentiation

Consider the curve with equation $x^3 + y^3 = 4xy$.

- (a) Use implicit differentiation to show that $\frac{dy}{dx} = \frac{4y - 3x^2}{3y^2 - 4x}$. [3]

The tangent to this curve is parallel to the x -axis at the point where $x = k$, $k > 0$.

- (b) Find the value of k . [5]

.....

.....

.....

.....

.....

.....

.....

.....

.....

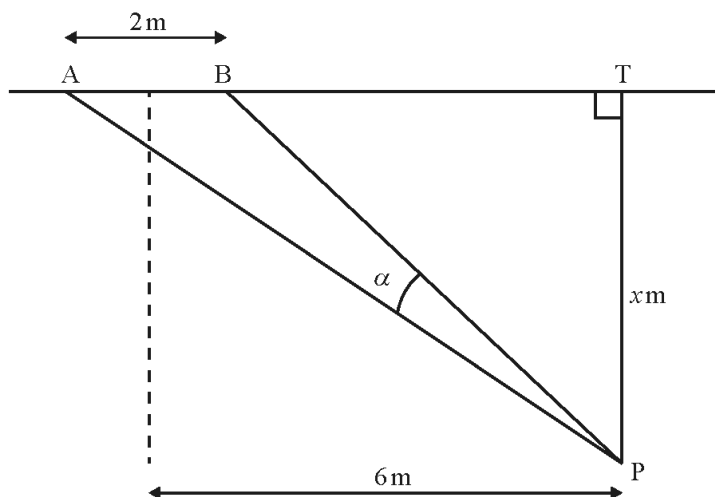
.....

.....



21-(MAT/22_HL_Summer_2016_Q11)-Differentiation

Points A, B and T lie on a line on an indoor soccer field. The goal, [AB], is 2 metres wide. A player situated at point P kicks a ball at the goal. [PT] is perpendicular to (AB) and is 6 metres from a parallel line through the centre of [AB]. Let PT be x metres and let $\alpha = \hat{APB}$ measured in degrees. Assume that the ball travels along the floor.



(a) Find the value of α when $x = 10$. [4]

(b) Show that $\tan \alpha = \frac{2x}{x^2 + 35}$. [4]

The maximum for $\tan \alpha$ gives the maximum for α .

(c) (i) Find $\frac{d}{dx}(\tan \alpha)$. [11]

(ii) Hence or otherwise find the value of α such that $\frac{d}{dx}(\tan \alpha) = 0$.

(iii) Find $\frac{d^2}{dx^2}(\tan \alpha)$ and hence show that the value of α never exceeds 10° . [11]

(d) Find the set of values of x for which $\alpha \geq 7^\circ$. [3]



22-(MAT/21_HL_Summer_2016_Q12)-Differentiation

Consider the curve, C defined by the equation $y^2 - 2xy = 5 - e^x$. The point A lies on C and has coordinates $(0, a)$, $a > 0$.

- (a) Find the value of a . [2]
- (b) Show that $\frac{dy}{dx} = \frac{2y - e^x}{2(y - x)}$. [4]
- (c) Find the equation of the normal to C at the point A. [3]
- (d) Find the coordinates of the second point at which the normal found in part (c) intersects C . [4]
- (e) Given that $v = y^3$, $y > 0$, find $\frac{dv}{dx}$ at $x = 0$. [3]



23-(MAT/22_HL_Summer_2016_Q12)-Differentiation, Integration

The functions f and g are defined by

$$f(x) = \frac{e^x + e^{-x}}{2}, x \in \mathbb{R}$$

$$g(x) = \frac{e^x - e^{-x}}{2}, x \in \mathbb{R}$$

- (a) (i) Show that $\frac{1}{4f(x) - 2g(x)} = \frac{e^x}{e^{2x} + 3}$.
- (ii) Use the substitution $u = e^x$ to find $\int_0^{\ln 3} \frac{1}{4f(x) - 2g(x)} dx$. Give your answer in the form $\frac{\pi\sqrt{a}}{b}$ where $a, b \in \mathbb{Z}^+$. [9]

Let $h(x) = nf(x) + g(x)$ where $n \in \mathbb{R}$, $n > 1$.

- (b) (i) By forming a quadratic equation in e^x , solve the equation $h(x) = k$, where $k \in \mathbb{R}^+$.
- (ii) Hence or otherwise show that the equation $h(x) = k$ has two real solutions provided that $k > \sqrt{n^2 - 1}$ and $k \in \mathbb{R}^+$. [8]

Let $t(x) = \frac{g(x)}{f(x)}$.

- (c) (i) Show that $t'(x) = \frac{[f(x)]^2 - [g(x)]^2}{[f(x)]^2}$ for $x \in \mathbb{R}$.
- (ii) Hence show that $t'(x) > 0$ for $x \in \mathbb{R}$. [6]



24–(MAT/20_HL_Winter_2016_Q6)–Differentiation

An earth satellite moves in a path that can be described by the curve $72.5x^2 + 71.5y^2 = 1$ where $x = x(t)$ and $y = y(t)$ are in thousands of kilometres and t is time in seconds.

Given that $\frac{dx}{dt} = 7.75 \times 10^{-5}$ when $x = 3.2 \times 10^{-3}$, find the possible values of $\frac{dy}{dt}$.

Give your answers in standard form.



25–(MAT/20_HL_Winter_2016_Q10)–Functions-Roots,Differentiation,Integration

Let the function f be defined by $f(x) = \frac{2-e^x}{2e^x-1}$, $x \in D$.

- (a) Determine D , the largest possible domain of f . [2]
- (b) Show that the graph of f has three asymptotes and state their equations. [5]
- (c) Show that $f'(x) = -\frac{3e^x}{(2e^x-1)^2}$. [3]
- (d) Use your answers from parts (b) and (c) to justify that f has an inverse and state its domain. [4]
- (e) Find an expression for $f^{-1}(x)$. [4]
- (f) Consider the region R enclosed by the graph of $y = f(x)$ and the axes. Find the volume of the solid obtained when R is rotated through 2π about the y -axis. [4]



26—(MAT/21_HL_Summer_2017_Q2)—Differentiation

The curve C is defined by equation $xy - \ln y = 1, y > 0$.

- (a) Find $\frac{dy}{dx}$ in terms of x and y . [4]
- (b) Determine the equation of the tangent to C at the point $\left(\frac{2}{e}, e\right)$. [3]



27–(MAT/22_HL_Summer_2017_Q2)–Differentiation, Integration

Consider the curve defined by the equation $4x^2 + y^2 = 7$.

- (a) Find the equation of the normal to the curve at the point $(1, \sqrt{3})$. [6]
- (b) Find the volume of the solid formed when the region bounded by the curve, the x -axis for $x \geq 0$ and the y -axis for $y \geq 0$ is rotated through 2π about the x -axis. [3]



28-(MAT/21_HL_Summer_2017_Q8)-Differentiation

A water trough which is 10 metres long has a uniform cross-section in the shape of a semicircle with radius 0.5 metres. It is partly filled with water as shown in the following diagram of the cross-section. The centre of the circle is O and the angle KOL is θ radians.

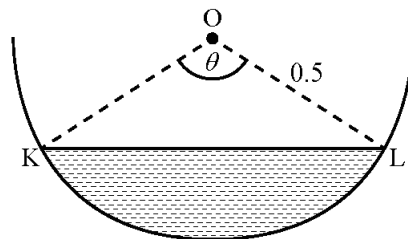


diagram not to scale

- (a) Find an expression for the volume of water $V(\text{m}^3)$ in the trough in terms of θ . [3]

The volume of water is increasing at a constant rate of $0.0008 \text{ m}^3 \text{ s}^{-1}$.

- (b) Calculate $\frac{d\theta}{dt}$ when $\theta = \frac{\pi}{3}$. [4]



29–(MAT/21_HL_Summer_2017_Q12)–Functions-Roots, Graphs, Differentiation

Consider $f(x) = -1 + \ln(\sqrt{x^2 - 1})$.

- (a) Find the largest possible domain D for f to be a function. [2]

The function f is defined by $f(x) = -1 + \ln(\sqrt{x^2 - 1})$, $x \in D$.

- (b) Sketch the graph of $y = f(x)$ showing clearly the equations of asymptotes and the coordinates of any intercepts with the axes. [3]
- (c) Explain why f is an even function. [1]
- (d) Explain why the inverse function f^{-1} does not exist. [1]

The function g is defined by $g(x) = -1 + \ln(\sqrt{x^2 - 1})$, $x \in]1, \infty[$.

- (e) Find the inverse function g^{-1} and state its domain. [4]
- (f) Find $g'(x)$. [3]
- (g) Hence, show that there are no solutions to
- (i) $g'(x) = 0$;
- (ii) $(g^{-1})'(x) = 0$. [4]



30–(MAT/20_HL_Winter_2017_Q10)–Differentiation, Integration

[Maximum mark: 17]

Consider the function $f(x) = \frac{\sqrt{x}}{\sin x}$, $0 < x < \pi$.

- (a) (i) Show that the x -coordinate of the minimum point on the curve $y = f(x)$ satisfies the equation $\tan x = 2x$.
- (ii) Determine the values of x for which $f(x)$ is a decreasing function. [7]
- (b) Sketch the graph of $y = f(x)$ showing clearly the minimum point and any asymptotic behaviour. [3]
- (c) Find the coordinates of the point on the graph of f where the normal to the graph is parallel to the line $y = -x$. [4]

Consider the region bounded by the curve $y = f(x)$, the x -axis and the lines $x = \frac{\pi}{6}$, $x = \frac{\pi}{3}$.

- (d) This region is now rotated through 2π radians about the x -axis. Find the volume of revolution. [3]



31–(MAT/20_HL_Winter_2017_Q11)–Trigonometry, Differentiation

[Maximum mark: 18]

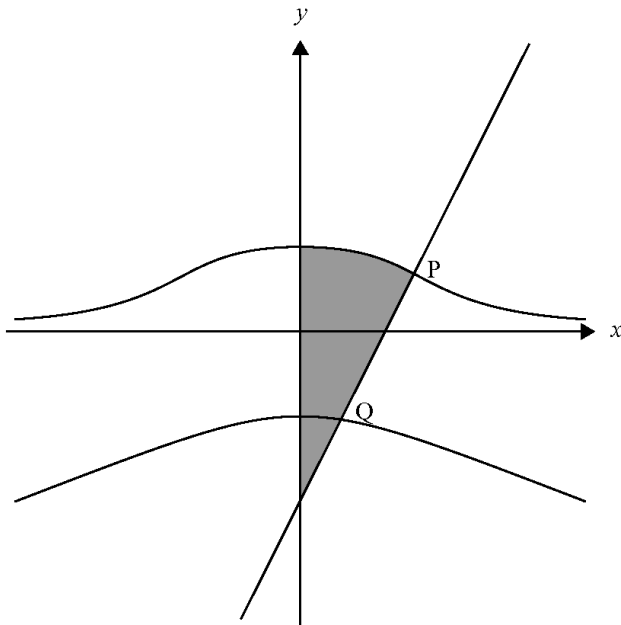
Consider the function $f(x) = 2\sin^2 x + 7\sin 2x + \tan x - 9$, $0 \leq x < \frac{\pi}{2}$.

- (a) (i) Determine an expression for $f'(x)$ in terms of x .
- (ii) Sketch a graph of $y = f'(x)$ for $0 \leq x < \frac{\pi}{2}$.
- (iii) Find the x -coordinate(s) of the point(s) of inflexion of the graph of $y = f(x)$, labelling these clearly on the graph of $y = f'(x)$. [8]
- (b) Let $u = \tan x$.
- (i) Express $\sin x$ in terms of u .
- (ii) Express $\sin 2x$ in terms of u .
- (iii) Hence show that $f(x) = 0$ can be expressed as $u^3 - 7u^2 + 15u - 9 = 0$. [7]
- (c) Solve the equation $f(x) = 0$, giving your answers in the form $\arctan k$ where $k \in \mathbb{Z}$. [3]



32-(MAT/21_HL_Summer_2018_Q9)-Differentiation,Integration

The following graph shows the two parts of the curve defined by the equation $x^2y = 5 - y^4$, and the normal to the curve at the point P(2, 1).



- (a) Show that there are exactly two points on the curve where the gradient is zero. [7]
- (b) Find the equation of the normal to the curve at the point P. [5]
- (c) The normal at P cuts the curve again at the point Q. Find the x-coordinate of Q. [3]
- (d) The shaded region is rotated by 2π about the y-axis. Find the volume of the solid formed. [7]



33-(MAT/22_HL_Summer_2018_Q11)-Differentiation

A curve C is given by the implicit equation $x + y - \cos(xy) = 0$.

(a) Show that $\frac{dy}{dx} = -\left(\frac{1 + y \sin(xy)}{1 + x \sin(xy)}\right)$. [5]

(b) The curve $xy = -\frac{\pi}{2}$ intersects C at P and Q.

(i) Find the coordinates of P and Q.

(ii) Given that the gradients of the tangents to C at P and Q are m_1 and m_2 respectively, show that $m_1 \times m_2 = 1$. [7]

(c) Find the coordinates of the three points on C , nearest the origin, where the tangent is parallel to the line $y = -x$. [7]



34–(MAT/20_HL_Winter_2018_Q5)–*Differentiation*

[Maximum mark: 5]

Differentiate from first principles the function $f(x) = 3x^3 - x$.



35–(MAT/20_HL_Winter_2018_Q9)–Functions-Roots,Differentiation

The function f is defined by $f(x) = \frac{2 \ln x + 1}{x - 3}$, $0 < x < 3$.

- (a) Find $f'(x)$. [4]
- (b) Hence, or otherwise, find the coordinates of the point of inflexion on the graph of $y = f(x)$. [4]
- (c) Draw a set of axes showing x and y values between -3 and 3 . On these axes
- (i) sketch the graph of $y = f(x)$, showing clearly any axis intercepts and giving the equations of any asymptotes.
- (ii) sketch the graph of $y = f^{-1}(x)$, showing clearly any axis intercepts and giving the equations of any asymptotes. [8]
- (d) Hence, or otherwise, solve the inequality $f(x) > f^{-1}(x)$. [3]



36—(MAT/21_HL_Summer_2019_Q1)—Differentiation

[Maximum mark: 4]

Let l be the tangent to the curve $y = xe^{2x}$ at the point $(1, e^2)$.

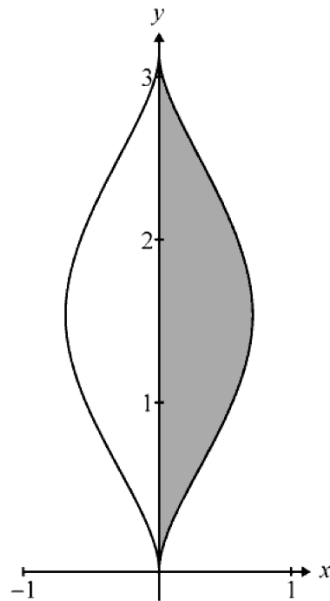
Find the coordinates of the point where l meets the x -axis.



37-(MAT/20_HL_Winter_2019_Q11)-Differentiation, Integration

[Maximum mark: 17]

The following diagram shows part of the graph of $2x^2 = \sin^3 y$ for $0 \leq y \leq \pi$.



- (a) (i) Using implicit differentiation, find an expression for $\frac{dy}{dx}$.
- (ii) Find the equation of the tangent to the curve at the point $\left(\frac{1}{4}, \frac{5\pi}{6}\right)$. [8]

The shaded region R is the area bounded by the curve, the y -axis and the lines $y = 0$ and $y = \pi$.

- (b) Find the area of R . [3]

The region R is now rotated about the y -axis, through 2π radians, to form a solid.

- (c) By writing $\sin^3 y$ as $(1 - \cos^2 y) \sin y$, show that the volume of the solid formed is $\frac{2\pi}{3}$. [6]



38–(MAT/20_HL_Winter_2020_Q8)–Differentiation

A small bead is free to move along a smooth wire in the shape of the curve $y = \frac{10}{3 - 2e^{0.5x}}$ ($x \geq 0$).

- (a) Find an expression for $\frac{dy}{dx}$. [3]

At the point on the curve where $x = 4$, it is given that $\frac{dy}{dt} = -0.1 \text{ m s}^{-1}$.

- (b) Find the value of $\frac{dx}{dt}$ at this exact same instant. [3]



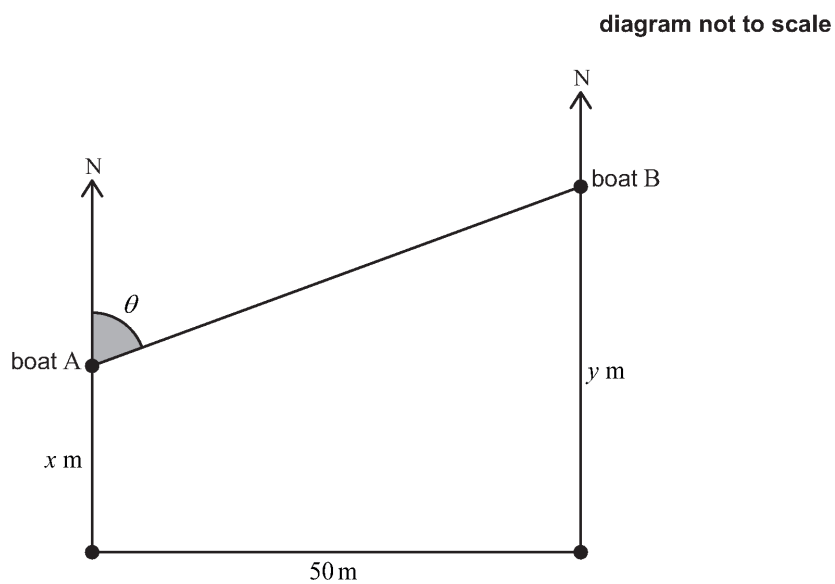
39–(MathAA/21_HL_Summer_2021_Q9)–Differentiation, Kinematics, Relative Velocity

[Maximum mark: 7]

Two boats A and B travel due north.

Initially, boat B is positioned 50 metres due east of boat A.

The distances travelled by boat A and boat B, after t seconds, are x metres and y metres respectively. The angle θ is the radian measure of the bearing of boat B from boat A. This information is shown on the following diagram.



- (a) Show that $y = x + 50 \cot \theta$. [1]

At time T , the following conditions are true.

Boat B has travelled 10 metres further than boat A.
Boat B is travelling at double the speed of boat A.
The rate of change of the angle θ is -0.1 radians per second.

- (b) Find the speed of boat A at time T . [6]



40–(MathAA/21_HL_Summer_2021_Q10)–Exponentials-Logarithms, Differentiation

[Maximum mark: 15]

Consider the function f defined by $f(x) = 90e^{-0.5x}$ for $x \in \mathbb{R}^+$.

The graph of f and the line $y = x$ intersect at point P.

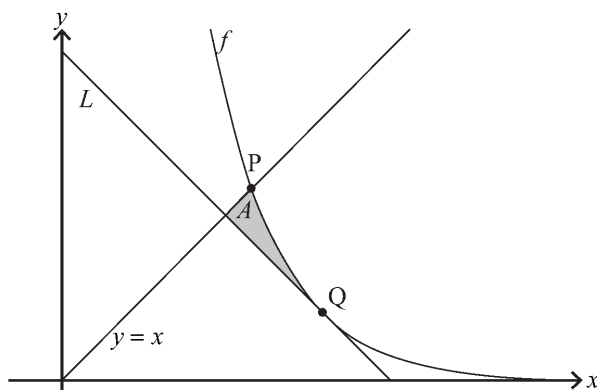
- (a) Find the x -coordinate of P. [2]

The line L has a gradient of -1 and is a tangent to the graph of f at the point Q.

- (b) Find the exact coordinates of Q. [4]

- (c) Show that the equation of L is $y = -x + 2\ln 45 + 2$. [2]

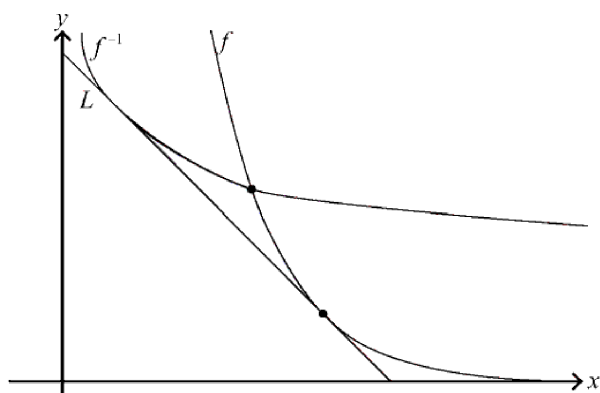
The shaded region A is enclosed by the graph of f and the lines $y = x$ and L .



- (d) (i) Find the x -coordinate of the point where L intersects the line $y = x$.
(ii) Hence, find the area of A . [5]



The line L is tangent to the graphs of both f and the inverse function f^{-1} .



(e) Find the shaded area enclosed by the graphs of f and f^{-1} and the line L .

[2]



41-(MathAA/22_HL_Summer_2021_Q11)-Differentiation

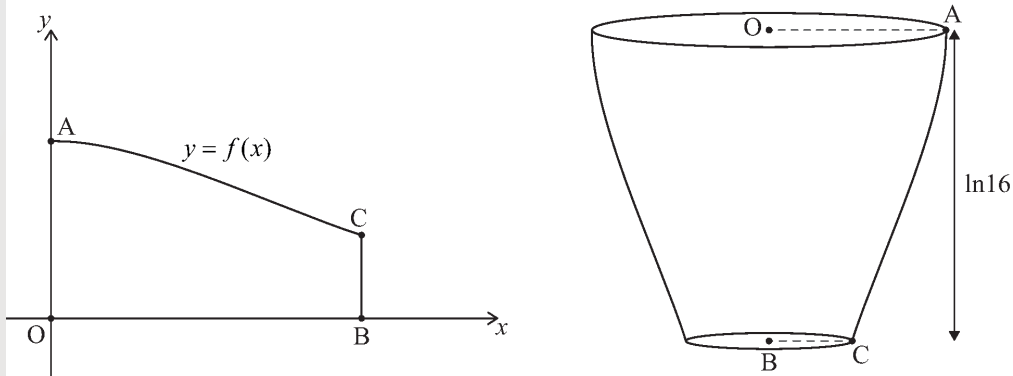
[Maximum mark: 18]

A function f is defined by $f(x) = \frac{ke^{\frac{x}{2}}}{1+e^x}$ where $x \in \mathbb{R}$, $x \geq 0$ and $k \in \mathbb{R}^+$.

The region enclosed by the graph of $y = f(x)$, the x -axis, the y -axis and the line $x = \ln 16$ is rotated 360° about the x -axis to form a solid of revolution.

- (a) Show that the volume of the solid formed is $\frac{15k^2\pi}{34}$ cubic units. [6]

Pedro wants to make a small bowl with a volume of 300 cm^3 based on the result from part (a). Pedro's design is shown in the following diagrams.



The vertical height of the bowl, BO , is measured along the x -axis. The radius of the bowl's top is OA and the radius of the bowl's base is BC . All lengths are measured in cm.

- (b) Find the value of k that satisfies the requirements of Pedro's design. [2]
- (c) Find
- OA ;
 - BC . [4]

For design purposes, Pedro investigates how the cross-sectional radius of the bowl changes.

- (d) (i) By sketching the graph of a suitable derivative of f , find where the cross-sectional radius of the bowl is decreasing most rapidly. [6]
- (ii) State the cross-sectional radius of the bowl at this point.



42–(MathAA/22_HL_Summer_2021_Q12)–Algebra-functions-Roots,Differentiation

[Maximum mark: 21]

A function f is defined by $f(x) = \arcsin\left(\frac{x^2-1}{x^2+1}\right)$, $x \in \mathbb{R}$.

(a) Show that f is an even function. [1]

(b) By considering limits, show that the graph of $y = f(x)$ has a horizontal asymptote and state its equation. [2]

(c) (i) Show that $f'(x) = \frac{2x}{\sqrt{x^2(x^2+1)}}$ for $x \in \mathbb{R}$, $x \neq 0$.

(ii) By using the expression for $f'(x)$ and the result $\sqrt{x^2} = |x|$, show that f is decreasing for $x < 0$. [9]

A function g is defined by $g(x) = \arcsin\left(\frac{x^2-1}{x^2+1}\right)$, $x \in \mathbb{R}$, $x \geq 0$.

(d) Find an expression for $g^{-1}(x)$, justifying your answer. [5]

(e) State the domain of g^{-1} . [1]

(f) Sketch the graph of $y = g^{-1}(x)$, clearly indicating any asymptotes with their equations and stating the values of any axes intercepts. [3]



43—(MathAA/20_HL_Winter_2021_Q8)—Differentiation

[Maximum mark: 8]

Consider the curve C given by $y = x - xy \ln(xy)$ where $x > 0$, $y > 0$.

- (a) Show that $\frac{dy}{dx} + \left(x \frac{dy}{dx} + y\right)(1 + \ln(xy)) = 1$. [3]
- (b) Hence find the equation of the tangent to C at the point where $x = 1$. [5]



44–(MathAA/21_HL_Summer_2022_Q10)–Algebra-functions-Roots,Differentiation,Integration

[Maximum mark: 20]

Consider the function $f(x) = \sqrt{x^2 - 1}$, where $1 \leq x \leq 2$.

(a) Sketch the curve $y = f(x)$, clearly indicating the coordinates of the endpoints. [2]

(b) (i) Show that the inverse function of f is given by $f^{-1}(x) = \sqrt{x^2 + 1}$.

(ii) State the domain and range of f^{-1} . [5]

The curve $y = f(x)$ is rotated 2π about the y -axis to form a solid of revolution that is used to model a water container.

(c) (i) Show that the volume, $V \text{ m}^3$, of water in the container when it is filled to a height of h metres is given by $V = \pi \left(\frac{1}{3} h^3 + h \right)$.

(ii) Hence, determine the maximum volume of the container. [5]

At $t = 0$, the container is empty. Water is then added to the container at a constant rate of $0.4 \text{ m}^3 \text{ s}^{-1}$.

(d) Find the time it takes to fill the container to its maximum volume. [2]

(e) Find the rate of change of the height of the water when the container is filled to half its maximum volume. [6]



45–(MathAA/22_HL_Summer_2022_Q10)–Trigonometry, Differentiation

[Maximum mark: 15]

A scientist conducted a nine-week experiment on two plants, A and B , of the same species. He wanted to determine the effect of using a new plant fertilizer. Plant A was given fertilizer regularly, while Plant B was not.

The scientist found that the height of Plant A , h_A cm, at time t weeks can be modelled by the function $h_A(t) = \sin(2t + 6) + 9t + 27$, where $0 \leq t \leq 9$.

The scientist found that the height of Plant B , h_B cm, at time t weeks can be modelled by the function $h_B(t) = 8t + 32$, where $0 < t < 9$.

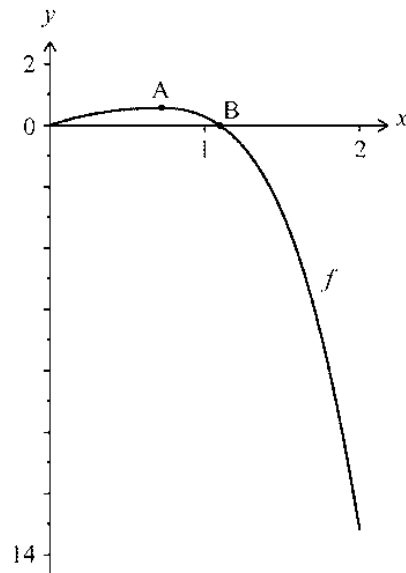
- (a) Use the scientist's models to find the initial height of
- (i) Plant B ;
 - (ii) Plant A correct to three significant figures. [3]
- (b) Find the values of t when $h_A(t) = h_B(t)$. [3]
- (c) For $t > 6$, prove that Plant A was always taller than Plant B . [3]
- (d) For $0 \leq t \leq 9$, find the total amount of time when the rate of growth of Plant B was greater than the rate of growth of Plant A . [6]



46–(MathAA/20_HL_Winter_2022_Q2)–Differentiation, Integration

[Maximum mark: 6]

The function f is defined as $f(x) = \ln(xe^x + 1) - x^4$, for $0 < x < 2$. The graph of f is shown in the following diagram.



The graph of f has a local maximum at point A. The graph intersects the x -axis at the origin and at point B.

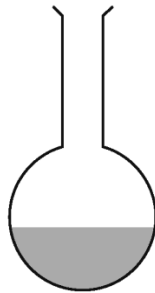
- (a) Find the coordinates of A. [2]
- (b) Find the x -coordinate of B. [1]
- (c) Find the total area enclosed by the graph of f , the x -axis and the line $x = 2$. [3]



47–(MathAA/20_HL_Winter_2022_Q8)–Differentiation

[Maximum mark: 6]

The following diagram shows liquid in a round-bottomed glass flask, which is made of a sphere and a cylindrical neck.



Initially, the flask is empty. Liquid is poured into the flask at a rate of $2\text{ cm}^3\text{ s}^{-1}$. You may assume that the liquid does not reach the cylindrical neck.

The volume $V\text{ cm}^3$ and the height $h\text{ cm}$ of the liquid in the flask satisfy the equation

$$V = 5\pi h^2 - \frac{1}{3}\pi h^3.$$

Find the rate of change of the height of the liquid in the flask at the instant when the volume of the liquid is 200 cm^3 .

